

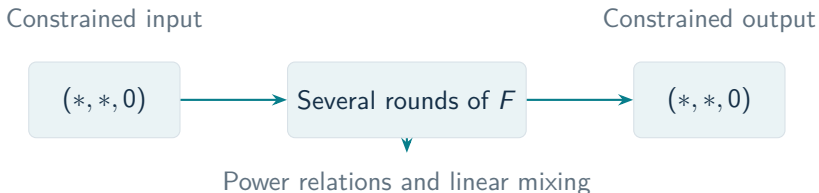
Efficient Polynomial System Solving via Dixon Resultants

Applications to AO Primitives

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AO cryptanalysis leads to polynomial systems

Arithmetic-friendly rounds admit compact algebraic descriptions.



CICO: find a state satisfying both endpoint constraints.

Introduce internal variables and solve the induced polynomial equations.

Paper, Section 2. CICO-1 schematic; stars denote free coordinates. AO means arithmetization-oriented.

Eliminated variables and retained parameters

$$\mathcal{F} = \{f_1, \dots, f_n\} \subseteq \mathbb{K}[\mathbf{a}][\mathbf{x}], \quad \mathbf{x} = (x_1, \dots, x_{n-1}).$$

Eliminate $n - 1$ variables

Combine the equations so that $\mathbf{x}$ no longer appears.

Retain $\mathbf{a} = (a_1, \dots, a_k)$

Produce a necessary relation $R(\mathbf{a}) = 0$.

When $k = 1$, the output is a univariate polynomial.

Its roots are candidates; recovery and verification finish the solving task.

Paper, Sections 2 and 5. A valid nonzero determinantal projection requires the stated applicability conditions.

Three algebraic routes

Method	Basic operation	Typical role
Gröbner bases	Transform the ideal into a useful basis	General system solving
Sylvester resultant	Two polynomials, one eliminated variable	Pairwise elimination
Dixon resultant	Several polynomials and eliminated variables	Simultaneous elimination

There is no universal winner. Structure, variable count, and degree matter.

Paper, Sections 1–3. GB solving may include an order change; this table does not restrict GB methods to one workflow.

The Dixon pipeline



$$\Delta(\mathbf{x}, \bar{\mathbf{x}}) = X\mathcal{M}\bar{X}^T.$$

X and $\bar{X}$ list monomials in two sets of variables.

Main cost drivers: matrix dimension, entry degrees, and structure.

Paper, Section 2.2. The maximal-rank-block route needs a valid projection condition for singular matrices.

Worked example: three equations in three variables

$$\begin{cases} f_1 = x + y - z = 0, \\ f_2 = x - y - 1 = 0, \\ f_3 = x^2 + y^2 - 5 = 0. \end{cases}$$

Eliminate: x, y .

Retain: z .

Goal: construct a polynomial $R(z)$ that every solution must satisfy.

Small enough to calculate by hand, using the actual Dixon construction.

Teaching example over $\mathbb{Q}$. The same calculation over $\mathbb{F}_{17}$ is given in the backup slides.

Step 1a: build the cancellation matrix

Introduce auxiliary variables u, v . Replace x , then y , one at a time.

$$C = \begin{bmatrix} x + y - z & x - y - 1 & x^2 + y^2 - 5 \\ u + y - z & u - y - 1 & u^2 + y^2 - 5 \\ u + v - z & u - v - 1 & u^2 + v^2 - 5 \end{bmatrix}.$$

$$(x, y) \xrightarrow{\text{replace } x} (u, y) \xrightarrow{\text{replace } y} (u, v)$$

Dixon construction as defined in the paper, instantiated on the teaching example.

Step 1b: remove the predictable difference factors

Use consecutive row differences, computed from the original rows:

$$\hat{C}_1 = C_1, \quad \hat{C}_2 = \frac{C_2 - C_1}{x - u}, \quad \hat{C}_3 = \frac{C_3 - C_2}{y - v}.$$

$$\hat{C} = \begin{bmatrix} x + y - z & x - y - 1 & x^2 + y^2 - 5 \\ -1 & -1 & -(x + u) \\ -1 & 1 & -(y + v) \end{bmatrix}.$$

$$\Delta = \det \hat{C} = \frac{\det C}{(x - u)(y - v)}.$$

For example, $(u^2 - x^2)/(x - u) = -(x + u)$. These are exact polynomial divisions, not numerical divisions at a chosen point.

Step 2: read off the Dixon matrix

Expanding the small determinant gives

$$\begin{aligned}\Delta = & 10 - (z + 1)x + (1 - z)y \\ & + (2x - z - 1)u + (2y - z + 1)v.\end{aligned}$$

Choose $X = (1, x, y)$ and $\bar{X} = (1, u, v)$. Then

$$\Delta = X \underbrace{\begin{bmatrix} 10 & -z-1 & 1-z \\ -z-1 & 2 & 0 \\ 1-z & 0 & 2 \end{bmatrix}}_{\mathcal{M}(z)} \bar{X}^T.$$

Teaching example. Each entry is the coefficient of one row monomial times one column monomial.

Steps 3–4: compute the eliminant and recover solutions

Our 3×3 matrix needs no submatrix extraction. Its determinant is

$$\begin{aligned} R(z) &= 40 - 2(z+1)^2 - 2(z-1)^2 \\ &= 36 - 4z^2 = -4(z^2 - 9). \end{aligned}$$

Find candidate roots

$z = 3$ or $z = -3$.

Recover and verify

$$x = (z+1)/2, \quad y = (z-1)/2.$$

$$(x, y, z) = (2, 1, 3), (-1, -2, -3).$$

The nonzero factor -4 does not affect roots over $\mathbb{Q}$. Both triples satisfy all three original equations.

Why is division by $(x - u)(y - v)$ exact?

Set $x = u$

The first two rows of C coincide.

Therefore $x - u$ divides $\det C$.

Set $y = v$

The last two rows of C coincide.

Therefore $y - v$ divides $\det C$.

These linear factors are relatively prime in the polynomial ring. Thus

$$\det C = (x - u)(y - v)\Delta, \quad \Delta \in \mathbb{Q}[x, y, u, v, z].$$

The general construction removes one predictable difference factor per eliminated variable.

Why does a common solution force $R(z) = 0$?

Let (x_0, y_0, z_0) satisfy the three original equations.

1. The first row of $C(x_0, y_0, u, v, z_0)$ is zero, so $\det C \equiv 0$.
2. The polynomial identity gives

$$(x_0 - u)(y_0 - v)\Delta(x_0, y_0, u, v, z_0) \equiv 0.$$

The first two factors are nonzero polynomials in u, v ; hence $\Delta \equiv 0$.

3. Comparing coefficients of $1, u, v$ yields

$$(1, x_0, y_0)\mathcal{M}(z_0) = 0.$$

The vector is nonzero. Therefore $\mathcal{M}(z_0)$ is singular and $\det \mathcal{M}(z_0) = 0$.

Polynomial cancellation is in the integral domain $\overline{\mathbb{Q}}[u, v]$. The evaluation vector includes the constant monomial.

The kernel vector is visible in the example

For $(x, y, z) = (2, 1, 3)$,

$$\mathcal{M}(3) = \begin{bmatrix} 10 & -4 & -2 \\ -4 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix}, \quad X(2, 1) = (1, 2, 1).$$

$$\begin{aligned} (1, 2, 1)\mathcal{M}(3) &= (10 - 8 - 2, -4 + 4, -2 + 2) \\ &= (0, 0, 0). \end{aligned}$$

A solution becomes a linear dependence in the coefficient matrix.

Teaching example. This is the concrete linear-algebra mechanism behind the eliminant.

A resultant root is a necessary condition

The converse can fail when degrees drop or extra factors appear.

Consider two polynomials in x , retaining t :

$$f_1 = tx - 1, \quad f_2 = tx - 2.$$

Their one-variable Dixon construction gives

$$\frac{\det \begin{bmatrix} tx - 1 & tx - 2 \\ tu - 1 & tu - 2 \end{bmatrix}}{x - u} = -t.$$

At $t = 0$ the eliminant vanishes, but the equations become $-1 = 0$ and $-2 = 0$.

Recover candidates and check them in the original system.

Teaching counterexample over $\mathbb{Q}$. Degree degeneration shows why determinant vanishing need not certify an affine solution.

Rectangular or singular Dixon matrices

What can happen?

The row and column supports have different sizes.

Or the matrix has generic rank $\rho < D$, even before parameter specialization.

Two separate checks: preserve rank, and preserve projection validity.

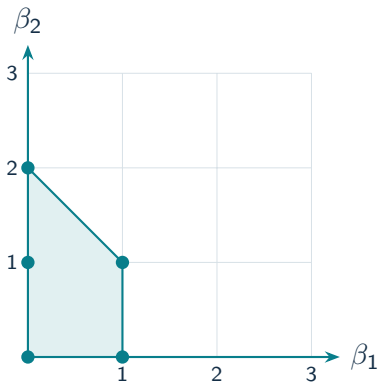
The KSY/RSC condition supplies a sufficient projection criterion. A nonzero minor alone is not enough.

Paper, Section 2.2 and Appendix F. A random specialization can lose rank; the singular-matrix route requires additional conditions.

Practical route

Evaluate at a generic point, identify independent rows and columns, and extract a $\rho \times \rho$ block.

Matrix size becomes a lattice-point count



Three quadratic equations

Monomial exponents obey

$$\beta_1 \leq 1, \quad \beta_1 + \beta_2 \leq 2.$$

Five feasible lattice points

Hence $D \leq 5$.

In higher dimensions, these constraints correspond to bounded lattice paths.

Paper, Section 3.1. Reverse-ordered exponents are shown. This quadratic illustration is separate from the mixed-degree worked example.

A sharper bound on the Dixon matrix dimension

For n equations of degree at most d in the eliminated variables,

$$D \leq C_n^{(d)} = \frac{1}{n(d-1)+1} \binom{nd}{n}.$$

Fuss–Catalan numbers count the allowed supports.

Unequal degrees lead to a Hessenberg determinant formula.

Example: $n = 5, d = 3$

Variable-wise bound

 1944

Volume comparison bound

 421

Refined support bound

 273

Paper, Section 3.1 and Appendix C. Generic attainment needs the characteristic hypothesis. Support dimension is not rank; these are bounds, not measured speedups.

Proof framework for the matrix-size bound

Upper bound

Difference
quotients
row i uses only
 $\bar{x}_1, \dots, \bar{x}_{i-1}$

Support lemma
 $\text{Supp}(\Delta_{\mathcal{F}}) \subseteq S$

Counting lemma
 $|S| = \text{bounded}$
lattice paths

Matrix-size bound
 $D(\mathcal{F}) \leq |S|$

Sharpness

Nested witness
 $f_1^* = 1$
 $f_i^* =$
 $(x_1 + \dots + x_{i-1})^{d_i}$

Triangularization
 $\Delta_{\mathcal{F}^*} = \prod_{i=2}^n \widehat{C}_{i,i}$

All terms survive
 $c_{\beta}(\mathcal{F}^*) \neq 0$
for every $\beta \in S$

Genericity lemma
one witness
 $\Rightarrow c_{\beta} \neq 0$
finite intersection
 U is dense open

$|S| = C_n^{(d)}$ in equal degree; unequal degrees give the Hessenberg formula.

Characteristic 0 or sufficiently large; support attainment does not assert full rank.

Paper, Section 3.1 and the proof of the support-bound lemma. Attainment concerns matrix support size, not full rank.

Complexity: size and degree both matter

For one retained parameter, the final polynomial matrix can be handled by HNF:

$$T_{\text{det}} = \tilde{O}(\rho^\omega E), \quad \rho \leq D, \quad E \leq nd.$$

Smaller matrix

Reducing ρ or its bound has a strong effect through ρ^ω .

Lower entry degree

Low-degree rows and columns can also reduce determinant work.

The full symbolic cost includes construction and rank extraction:

$$T_{\text{symbolic}} = \tilde{O}(\max\{T_{\text{construct}}, T_{\text{extract}}, T_{\text{det}}\}).$$

Paper, Sections 2.3 and 3.2. Here $E \geq 1$ bounds entry degree and input total degrees are at most d . Recovery is separate; construction cannot always be discarded.

Choosing a submatrix with lower degree

Favor low-degree rows and columns while preserving rank.

$$\deg \det A \leq \min \left\{ \sum \text{rdeg}(A), \sum \text{cdeg}(A) \right\}.$$

Observed degree test	Random	Degree-aware
Determinant degree exceeds the Bézout bound	963/1000	1/1000

A useful heuristic, with no guarantee of eliminating every extraneous factor.

Paper, Section 4.1 and Appendix G. Five configurations, 200 trials each, over $\mathbb{F}_{65537}$. Passing the test does not certify a factor-free eliminant.

DRSolve: putting the pipeline into practice

Implementation

Open-source C code built on FLINT and PML.

Prime fields, extension fields, and rational arithmetic.

Algorithm choices

Several matrix-construction and determinant routines.

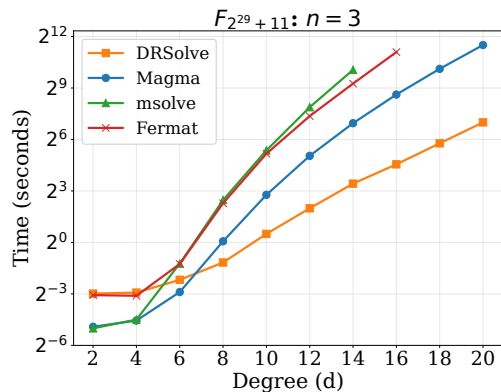
Degree-aware submatrix selection and complexity estimation.

Elimination and complete solving are distinct tasks.

A benchmark should specify which output and which stages are timed.

Paper, Section 4.2. github.com/drsolve/drsolve. Subsequent timings are the paper's reported measurements.

Fewer variables and higher degree favor Dixon



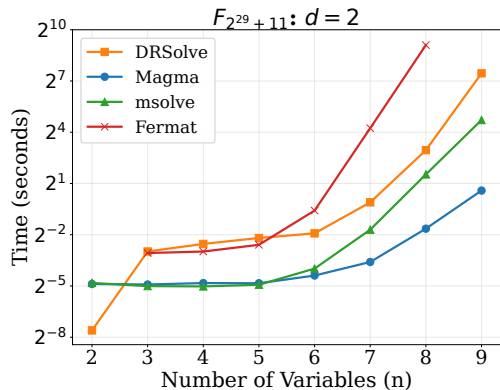
Three equations

Three variables

DRSolve is competitive as the degree increases in these tests.

Paper, Figure 2 ($n=3$ panel). Dense random systems over $\mathbb{F}_p$, $p = 2^{29} + 11$. Time in seconds on a log scale. Curves stop at the one-hour timeout.

More variables and low degree favor GB solvers



Quadratic systems

As the variable count grows, Magma and msolve remain more competitive.

Consider variable count and degree together.

Paper, Figure 2 ($d = 2$ panel). Equation count equals variable count; same field as the preceding slide.

Three elimination patterns for AO systems

Pattern	How it uses structure	Example
Direct	Eliminate from the complete square system	Poseidon
Iterative	Process local relations along a chain	Vision
Reduction + elimination	Use triangular relations before merging constraints	XHash12

The algebraic model determines how elimination should be organized.

Paper, Section 5. These organizing strategies do not all improve every parameter regime.

Poseidon: direct elimination and structural effects

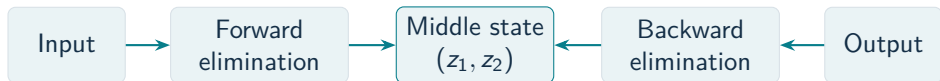
Some small instances favor Dixon; others favor GB.

Configuration (R_F, R_P, α)	GB time	Dixon time
(3, 0, 3)	2.2 s	0.3 s
(4, 0, 3)	188189 s	1772 s
(3, 1, 3)	178 s	695 s
(2, 2, 3)	0.4 s	11.9 s

Partial-round structure can reduce the practical cost of both methods.

Paper, Table 4, selected rows. $t = 6, c = 1, d = 3$, I/O modeling, bypass one round; $p = 2^{62} + 169$. R_F, R_P denote full and partial rounds.

Vision: meet in a two-variable middle state



$$A(z_1, z_2) = 0, \quad B(z_1, z_2) = 0 \quad \xrightarrow{\text{Res}_{z_1}} \quad R(z_2) = 0.$$

Each chain contributes one relation in the shared middle state.

The final merge dominates the paper's listed stage estimates, subject to its degree assumptions.

Paper, Section 5.2 and Appendix K, $s = 2$. The merge eliminates one of two middle variables. Actual determinantal output-degree bounds are conditional.

XHash12: reduce powers before eliminating

$$\begin{aligned} z_1^\alpha &= f_1(\mathbf{x}), \\ z_2^\alpha &= f_2(\mathbf{x}, z_1), \\ &\vdots \end{aligned}$$

Triangular relations replace high powers by reduced expressions.

The hybrid route

1. Reduce the output constraints.
2. Eliminate internal variables successively.
3. Merge the input relations with Dixon elimination.

Reduction lowers exponents; it does not itself eliminate all internal variables.

Paper, Section 5.3 and Appendix L. CICO-1 inherits the optimized pairwise-resultant route of prior work.

The hybrid route has its own trade-offs

Paper's symbolic comparison: strong for CICO-1, a small margin for CICO-2, and worse than direct elimination for CICO-3/4.

Reported small-instance elimination times

Instance (α , width, steps)	GB	Direct	Hybrid
CICO-1 (3, 3, 5)	1.5 s	5.12 s	0.04 s
CICO-2 (3, 4, 3)	59.7 s	4.5 s	413.2 s

A smaller generic cost estimate need not imply a faster small-instance implementation.

Paper, Figure 5 and Table 6, selected rows. Theoretical and measured comparisons are different kinds of evidence.

What is established, and what remains conditional?

Established structure

Explicit bounds on monomial support.

Generic attainment under the stated characteristic hypothesis.

An elimination implementation and benchmark results.

Remaining challenges

Support dimension can greatly exceed rank.

Extraneous-factor control is heuristic.

Structured cryptographic systems need more precise estimates.

Paper, Sections 3–6. Projection conditions, degree hypotheses, and recovery costs must remain explicit.

Takeaways

1. A common solution creates a kernel vector in the Dixon matrix.
2. Lattice-path counting sharpens the matrix-size bound.
3. Practical gains depend on variable count, degree, and system structure.

Thank you. Questions?

github.com/drsolve/drsolve

Backup: a nonzero minor alone is not enough

Consider the abstract polynomial matrix

$$A(t) = \begin{bmatrix} 1 & t \\ 0 & 0 \end{bmatrix}.$$

It has a nonzero kernel for every t , but a maximal-rank minor is the constant 1.

For a generically singular matrix, singularity itself says nothing new.

To make maximal minors vanish at solutions, one needs a drop below the generic rank. The Dixon KSY/RSC criterion supplies a sufficient condition in the setting used by the paper.

Illustrative linear-algebra example, not claimed to be a Dixon matrix. See the paper's Section 2.2 for the projection criterion.

Backup: the same example over $\mathbb{F}_{17}$

The polynomial identities hold after reduction modulo 17:

$$R(z) = -4(z^2 - 9), \quad z \in \{3, 14\}.$$

Since $2^{-1} = 9$ in $\mathbb{F}_{17}$,

$$x = 9(z + 1), \quad y = 9(z - 1) \pmod{17}.$$

z	x	y
3	2	1
14	16	15

These are all solutions in $\mathbb{F}_{17}^3$.

Teaching example. The source package includes exact identity checks and exhaustive verification over $\mathbb{F}_{17}$.

Backup: unequal degrees and the worked example

For sorted positive degree bounds $d_1 \leq \dots \leq d_n$, set

$$a_i = \sum_{j=1}^i (d_{n+1-j} - 1).$$

The paper bounds the support dimension by

$$D \leq \det_{1 \leq i, j \leq n-1} \left[\binom{a_i + 1}{j - i + 1} \right].$$

For $(d_1, d_2, d_3) = (1, 1, 2)$, we have $(a_1, a_2) = (1, 1)$:

$$D \leq \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 3.$$

Paper, Section 3.1. The worked example's 3×3 matrix attains this support-dimension bound. Generic support attainment does not assert full rank in general.

Backup: selected references

- ▶ H. Suo and J. Cui. *Efficient Polynomial System Solving via Dixon Resultants: Applications to AO Primitives*. Uploaded full version.
- ▶ A. L. Dixon (1909). Original multivariate resultant construction.
- ▶ D. Kapur, T. Saxena and L. Yang (1994). Dixon-based resultant algorithms and the singular-matrix setting.
- ▶ G. Labahn, V. Neiger and W. Zhou (2017). Fast univariate polynomial-matrix determinant and Hermite-form algorithms.

eprint.iacr.org/2026/1969 github.com/drsolve/drsolve

Full bibliographic details and the application-specific literature are in the uploaded paper.